Trypsin hydrolyses proteins into polypeptides and peptones, while chymotrypsin hydrolyses peptide bonds associated with specific amino acids.

The pancreatic amylase converts glycogen and starch into maltose. Lipase acts on emulsified fat (triglycerides) and hydrolyses them into free fatty acid and monoglycerides. Monoglycerides are further hydrolysed to fatty acid and glycerol. Nucleases in the pancreatic juice break the nucleic acid into nucleotides and nucleosides.

The secretions of the Brunner's gland along with the secretions of the intestinal glands constitute the intestinal juice or **succus entericus**. The enzymes in the intestinal juice such as maltase, lactase, sucrase (invertase), peptidases, lipases, nucleotidases and nucleosidases act on the breakdown products of bile and pancreatic digestion.

Maltose	Maltase →	glucose + glucose
Sucrose	Sucrase →	glucose + fructose
Lactose	Lactase →	glucose + galactose
Dipeptides, Tripeptides	Peptidase →	amino acids
Nucleotides	Nucleotidase →	Nucleoside + Phosphoric acid
Nucleoside	Nucleosidase →	Sugar + Nitrogen base
Diglycerides and monoglycerides	Lipases →	Fatty acids + glycerol

The mucus along with the bicarbonate ions from the pancreas provides an alkaline medium (pH 7.8) for the enzymatic action. As a result of digestion, all macromolecules of food are converted into their corresponding monomeric units.

Carbohydrates ⟶ monosaccharides (glucose, fructose, galactose)

Proteins ⟶ amino acids

Lipids ⟶ fatty acids and glycerol

The simple substances thus formed are absorbed in the jejunum and ileum region of the small intestine. The undigested and unabsorbed substances are propelled into the large intestine. The activities of the gastro-intestinal tract are carried out by the neural and hormonal control for proper coordination of different parts. Gastric and intestinal secretions are stimulated by neural signals. Hormonal control of the secretion of digestive juices is carried out by local hormones produced by the gastric and intestinal mucosa.

5.3 Absorption and assimilation of proteins, carbohydrates and fats

Absorption is a process by which the end product of digestion passes through the intestinal mucosa into the blood and lymph. The villi in the lumen of ileum are the absorbing units, consisting of a lacteal duct in the middle surrounded by fine network of blood capillaries. The process of absorption involves active, passive and facilitated transport. Small amounts of glucose, amino acids and electrolytes like chloride ions are generally absorbed by simple diffusion. The passage of these substances into the blood depends upon concentration gradients. However, some of the substances like fructose are absorbed with the help of the carrier ions like Na^+. This mechanism is called facilitated transport.

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